

CENTRAL REGISTRY OF RARE DISEASES (CRRD)

Data Quality Benchmark Report
(1st Edition: 2015 – 2025)

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Acknowledgments

The first edition of the Central Registry of Rare Diseases (CRRD) Data Quality Benchmark Report reflects the combined efforts of many individuals and institutions who have contributed to building and maintaining Belgium's national rare disease registry over the past decade. What began as an ambitious initiative to systematically collect rare disease data has evolved into a robust, coordinated data collection network that supports national monitoring, informs policy development, and enhances understanding of rare diseases.

We express our sincere appreciation to all former and current CRRD registry managers and data managers whose efforts have ensured the continuity and operational stability of the data collection process. Their dedication to maintaining and updating the data flows, supporting data providers, and implementing evolving Data Collection Definitions (DCD) has been instrumental to the registry's development.

Our deepest gratitude goes to the participating genetic centres whose data were analysed for this report. Their clinical, administrative, and data teams continue to invest significant effort in submitting accurate and timely data, often alongside demanding clinical responsibilities and evolving technical requirements. Their dedication is particularly appreciated given the voluntary nature of registration and the absence of any incentives. Their commitment forms the backbone of the CRRD and enables the continuous enrichment of the national dataset.

We would also like to acknowledge our institutional partners and collaborators, including the Federal Public Service Health, Food Chain Safety and Environment (FPS Public Health), the Healthdata.be team, and all stakeholders involved in designing, supporting, and maintaining the CRRD data collection framework. Their guidance and technical support ensure that the registry remains aligned with national standards and operational needs.

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Above all, we thank the patients and families whose information forms the heart of this registry. Their willingness to contribute their data plays a crucial role not only in advancing knowledge but also in strengthening the national response to rare diseases. We are equally grateful to RaDiOrg, Belgium's national rare disease patient organization, for their ongoing engagement, advocacy, and support in raising awareness and representing the broader rare disease community.

This report is the product of that collective commitment. We are grateful for the continued trust, collaboration, and shared dedication to advancing the understanding of rare diseases and improving patient outcomes in Belgium.

The CRRD team

EXECUTIVE SUMMARY

This first edition of the Data Quality Benchmark Report presents a global overview of the quality of rare disease data collected by seven accredited Belgian genetic centres between 2015 and 2025, under Architecture 1 of the Central Registry of Rare Diseases (CRRD).

The report comprises two complementary components:

1. An **aggregated global-level analysis** describing overall data quality trends across centres (Part I), and
2. A **confidential centre-specific benchmark** report providing **pseudonymised** comparative insights to support internal quality improvement and harmonization (Part II).

The objectives of this report are to:

- Evaluate the completeness, validity and reliability, consistency and disease-coding quality of CRRD data;
- Assess key indicators related to the CRRD data validation workflow;
- Identify areas of convergence and variability across genetic centres;
- Support harmonised registration practices through constructive, evidence-based feedback;
- Strengthen the reliability and comparability of national rare disease information for epidemiological and policy purposes.

At aggregated-global level, the CRRD dataset expanded steadily between 2015 and 2025, reaching **76 894 registrations** corresponding to **21 963 unique patients**, based on the **full set of raw submissions prior to validation and data cleaning**. This reflects the progressive onboarding of centres and consolidation of registration practices.

Data completeness was consistently high for identification (100,0%) and consultation-related variables (93,2%), while diagnostic information showed lower completeness (76,7%), reflecting the inherent complexity and heterogeneity of rare disease diagnosis.

Overall **validity and consistency indicators** demonstrated a high level of **data reliability**, with the vast majority of variables showing error rates below 1%. Variables requiring greater standardization—such as postcodes, age ranges, and disease coding—accounted for most observed validation issues. Throughout the reporting period, **ORPHAcodes** remained the dominant disease classification system, while SNOMED CT, OMIM, HPO, and ICD-10 were used either as standalone systems or in combination with ORPHAcodes, reflecting evolving coding practices and interoperability efforts.

Validation process indicators indicated a gradual improvement in registration efficiency over time, as reflected by increasing first-attempt validation rates and progressive alignment with successive versions of the Data Collection Definition (DCD). At the same time, repeated resubmissions highlight both rigorous adherence to validation rules and the associated workload for data providers.

While **aggregated-global trends demonstrate overall convergence and gradual maturation of CRRD data quality, variability across centres persists in specific areas**, including diagnostic coding strategies, use of ‘unknown’ values, and completeness of selected mandatory fields. These differences are explored in detail in the **centre-specific benchmark reports** and should be interpreted in light of the primary coding classification system used at each centre, local workflows, data availability in electronic health records (EHRs), resource constraints, and the clinical heterogeneity and complexity of rare diseases. Additionally, inherent DCD specifications and known technical limitations should be considered.

The findings of this report are intended to foster constructive dialogue, guide quality-improvement actions and support the continued development of a robust national rare disease surveillance system.

RÉSUMÉ FRANÇAIS

Cette première édition du Rapport de benchmark sur la qualité des données présente une vue d'ensemble de la qualité des données relatives aux maladies rares collectées par sept centres belges de génétique agréés entre 2015 et 2025, dans le cadre de l'Architecture 1 du Registre Central des Maladies Rares (CRRD).

Le rapport se compose de deux volets complémentaires :

1. **Une analyse agrégée au niveau global** décrivant les tendances générales de la qualité des données à travers les centres (Partie I), et
2. Un **rapport de benchmarking confidentiel spécifique à chaque centre**, fournissant des analyses comparatives **pseudonymisées** afin de soutenir l'amélioration interne de la qualité et l'harmonisation des pratiques (Partie II).

Les objectifs de ce rapport sont les suivants :

- Évaluer l'exhaustivité, la validité et la fiabilité, la cohérence et la qualité des données du CRRD ;
- Identifier les domaines de convergence et de variabilité entre les centres de génétique ;
- Soutenir des pratiques d'enregistrement harmonisées au moyen d'un retour constructif et fondé sur des données probantes ;
- Renforcer la fiabilité et la comparabilité des informations nationales relatives aux maladies rares à des fins épidémiologiques et de politiques publiques.

Au niveau agrégé global, le jeu de données du CRRD s'est progressivement élargi entre 2015 et 2025, atteignant **76 894 enregistrements** correspondant à **21 963 patients uniques**, sur base de **l'ensemble des soumissions brutes reçues avant toute validation et nettoyage des données**. Cette évolution reflète l'intégration progressive des centres et la consolidation des pratiques d'enregistrement.

L'**exhaustivité des données** était systématiquement élevée pour les variables d'identification (100,0 %) et celles liées aux consultations (93,2 %), tandis que les informations diagnostiques présentaient une exhaustivité plus faible (76,7 %), traduisant la complexité et l'hétérogénéité intrinsèques du diagnostic des maladies rares.

Les indicateurs globaux de **validité et de cohérence** témoignent d'un niveau élevé de **fiabilité des données**, la grande majorité des variables présentant des taux d'erreur inférieurs à 1 %. Les variables nécessitant un degré de standardisation plus élevé — telles que les codes postaux, les classes d'âge et le codage des maladies — concentrent la majorité des problèmes de validation observés. Sur l'ensemble de la période analysée, les **ORPHAcodes** sont demeurés le système de classification des maladies prédominant, tandis que SNOMED CT, OMIM, HPO et la ICD-10 ont été utilisés soit comme systèmes autonomes, soit en combinaison avec les ORPHAcodes, reflétant l'évolution des pratiques de codage et les efforts en matière d'interopérabilité.

Les **résultats de la validation** indiquent une amélioration progressive de l'efficacité de l'enregistrement au fil du temps, mise en évidence par l'augmentation des taux de validation dès la première soumission et par l'alignement progressif sur les versions successives de la Définition de la Collecte de Données (DCD). Parallèlement, les resoumissions répétées traduisent à la fois une application rigoureuse des règles de validation et la charge de travail associée pour les fournisseurs de données.

Si **les tendances agrégées globales montrent une convergence générale et une maturation progressive de la qualité des données du CRRD, des variations entre centres persistent dans certains domaines**, notamment les stratégies de codage diagnostique, l'utilisation des valeurs « inconnu » et l'exhaustivité de certains champs obligatoires. Ces différences sont examinées en détail dans les rapports de **benchmarking spécifiques à chaque centre** et doivent être interprétées à la lumière du système de classification principal utilisé par chaque centre, des flux de travail locaux, de la disponibilité des données dans les dossiers médicaux électroniques, des contraintes en ressources, ainsi que de l'hétérogénéité clinique et de la complexité des maladies rares. En outre, les spécifications intrinsèques de la DCD et certaines limitations techniques connues doivent également être prises en compte.

Les résultats de ce rapport visent à favoriser un dialogue constructif, à orienter les actions d'amélioration de la qualité et à soutenir le développement continu d'un système national robuste de surveillance des maladies rares.

NEDERLANDSTALIGE SAMENVATTING

Deze eerste editie van het **Datakwaliteit Benchmark Rapport** biedt een globaal overzicht van de kwaliteit van gegevens over zeldzame ziekten die tussen 2015 en 2025 werden verzameld door zeven geaccrediteerde Belgische genetische centra, binnen Architectuur 1 van het Centraal Register Zeldzame Ziekten (CRRD).

Het rapport bestaat uit twee complementaire onderdelen:

1. Een **geaggregeerde analyse op globaal niveau** waarin de algemene trends in datakwaliteit over de centra heen worden beschreven (Deel I), en
2. Een **vertrouwelijk, centrumspecifiek benchmarkrapport** dat **gepseudonimiseerde** vergelijkende inzichten biedt ter ondersteuning van interne kwaliteitsverbetering en harmonisatie (Deel II).

De doelstellingen van dit rapport zijn:

- Het evalueren van de volledigheid, validiteit en betrouwbaarheid, consistentie en kwaliteit van CRRD-gegevens;
- Het identificeren van overeenkomsten en variabiliteit tussen de genetische centra;
- Het ondersteunen van geharmoniseerde registratiepraktijken via constructieve, evidence-based feedback;
- Het versterken van de betrouwbaarheid en vergelijkbaarheid van nationale informatie over zeldzame ziekten voor epidemiologische en beleidsdoeleinden.

Op geaggregeerd globaal niveau kende de CRRD-dataset tussen 2015 en 2025 een gestage groei tot **76 894 registraties**, overeenkomend met **21 963 unieke patiënten**, gebaseerd op **alle aangeleverde gegevens vóór toepassing van validatie- en datacleaningprocedures**. Deze evolutie weerspiegelt de geleidelijke toetreding van centra en de consolidatie van registratiepraktijken.

De volledigheid van de gegevens was consistent hoog voor identificatie- (100,0%) en consultatiegerelateerde variabelen (93,2%). Daarentegen vertoonde de diagnostische informatie een lagere volledigheid (76,7%), wat de inherente complexiteit en heterogeniteit van de diagnostiek van zeldzame ziekten weerspiegelt.

Indicatoren voor **validiteit en consistentie** wezen op een hoog niveau van **databetrouwbaarheid**, waarbij voor de overgrote meerderheid van de variabelen foutpercentages onder 1% werden vastgesteld. Variabelen die een hogere mate van standaardisatie vereisen—zoals postcodes, leeftijdscategorieën en ziektecodering—waren verantwoordelijk voor het merendeel van de geobserveerde validatieproblemen. Tijdens de volledige rapporteringsperiode bleven **ORPHAcodes** het dominante ziekteclassificatiesysteem. SNOMED CT, OMIM, HPO en ICD-10 werden gebruikt als afzonderlijke systemen of in combinatie met ORPHAcodes, wat wijst op evoluerende coderingspraktijken en toenemende inspanningen in functie van interoperabiliteit.

Validatie-uitkomsten wezen op een geleidelijke verbetering van de registratie-efficiëntie in de loop van de tijd, zoals blijkt uit toenemende eerste-poging validatiepercentages en progressieve afstemming op opeenvolgende versies van de Data Collection Definition (DCD). Tegelijkertijd benadrukken herhaaldelijk ingediende registraties zowel de strikte naleving van validatieregels als de bijbehorende werklast voor gegevensleveranciers.

Hoewel **geaggregeerde globale trends een algemene overeenkomst en een geleidelijke maturiteit van de kwaliteit van CRRD-gegevens aantonen, blijft er variabiliteit tussen de centra bestaan in specifieke domeinen**, zoals diagnostische coderingsstrategieën, het gebruik van 'onbekende' opties en de volledigheid van geselecteerde verplichte velden. Deze verschillen worden in detail besproken in de **centrumspecifieke benchmarkrapporten** en dienen te worden geïnterpreteerd in het licht van het primaire classificatiesysteem dat in elk centrum wordt gebruikt, lokale workflows, beschikbaarheid van gegevens in het elektronisch patiëntendossier, beperkte middelen, en de klinische heterogeniteit en complexiteit van zeldzame ziekten. Daarnaast moeten ook de inherente specificaties van de DCD en gekende technische beperkingen in beschouwing worden genomen.

De bevindingen van dit rapport zijn bedoeld om een constructieve dialoog te stimuleren, kwaliteitsverbeterings-initiatieven te begeleiden en de verdere ontwikkeling van een robuust nationaal surveillancesysteem voor zeldzame ziekten te ondersteunen.

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LIST OF ACRONYMS AND ABBREVIATIONS

BE	Belgium
CRRD	Central Registry of Rare Diseases
CSV	Comma-Separated Values
DCD	Data Collection Definition
DWH	Data Warehouse
EHR	Electronic Health Record
FPS Public Health	Federal Public Service Health, Food Chain Safety and Environnement
HD4DP	Healthdata for Data Provider
HD4RES	Healthdata for Researchers
HGVS	Human Genome Variation Society (Nomenclature)
HMIS	Health Management Information System
HPO	Human Phenotype Ontology
ICD-10	International Classification of Diseases, 10 th Revision
ICD-O	International Classification of Diseases for Oncology
NIHDI	National Institute for Health and Disability Insurance
ISCN	International System for Human Cytogenomic Nomenclature
LIMS	Laboratory Information Management System
NISS	National Registration Number
OMIM	Online Mendelian Inheritance in Man
POC	Proof of Concept
RDFs	Rare Disease Functions
SAS	Statistical Analysis Software
SCRA	Small Cell Risk Assessment

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PART I

Global Data Quality Overview of the CRRD

1. INTRODUCTION

1.1. REPORT STRUCTURE

This CRRD data quality benchmark report is organised in two complementary components:

Part I: Global Data Quality Overview

This part provides aggregated data quality indicators across all contributing genetic centres. It offers a global view of completeness, validity and reliability, consistency, disease-coding quality and validation process indicators, and is accessible to all stakeholders involved in rare disease policy, research, health service planning, and patient organisations.

Part II: Centre-Specific Data Quality Benchmark

Each participating genetic centre receives an individual, confidential report containing its own detailed data quality indicators and pseudonymised comparisons to global trends. These tailored insights are intended to support internal data quality monitoring and continuous improvement, without enabling inter-centre identification. A comparative analysis of clinical and organisational workflows between centres falls outside the intended scope of this report.

1.2. BACKGROUND AND OBJECTIVES

The CRRD was developed by Sciensano to provide a national framework for the standardised registration and monitoring of rare diseases in Belgium. Its aim is to consolidate information collected from accredited genetic centres and Rare Disease Functions (RDFs) into a single, harmonised national database that supports surveillance, research, health service planning and policy evaluation.

This report presents a global assessment of data quality and completeness for all patient registrations submitted under Architecture 1 of the CRRD in 2015–2025, corresponding to versions v1–v3 implemented through Healthdata.be. During this period, only the genetic centres¹ contributed data to the registry. The RDFs were onboarded later, in September 2024, under Architecture 2 (v5), and will be included in future reporting cycles. Their data are therefore not part of this analysis.

The figures presented in this report reflect the **raw, uncleaned registrations** received through Architecture 1, including both initial submissions as well as resubmissions generated through the validation feedback loop. They do not represent the final set of validated registrations stored in the Data Warehouse (DWH) of the CRRD. For cleaned, validated, and fully quality-checked data, please refer to the CRRD Data Collection Report².

1.3. SCOPE OF PART I

Part I of this report provides a global assessment of data quality for all registrations submitted under Architecture 1 of the CRRD. Its objectives are to:

- Analyse global trends over time in registry submissions (2015–2025);

¹ These centres correspond to the accredited genetic centres in Belgium, as defined under the Royal Decree of 2008 establishing the network of medical genetics services.

² Dhondt E, Jirari A, Demeulemeester N, Wuyts J. Central Registry of Rare Diseases (CRRD), Data Collection Report (1st Edition 2015–2025). Brussels, Belgium: Sciensano; 2026. 97 p. Report reference number: D/2026.14.440/6.

- Evaluate the completeness, validity and reliability, consistency and disease-coding quality of CRRD data;
- Assess key indicators related to the CRRD data validation workflow;
- Identify areas of convergence and variability across genetic centres;
- Highlight common data quality challenges related to data capture, validation, and disease coding;
- Provide an evidence base to support harmonised registration practices and continuous improvement of data entry and validation processes.

In this report, the term **data capture** refers to the availability and recording of information at the source level (e.g., EHRs), whereas **data entry** refers to the subsequent submission or transcription of this information into the CRRD through the registry's data submission tools. The technical aspects of the data flow and submission process are described in Section 2.3.

In parallel, a separate centre-specific benchmark report is produced for each genetic centre (Part II). These confidential reports provide detailed evaluations and pseudonymised peer-comparison indicators to support centre-level data quality improvement.

1.4. STRUCTURE OF PART I

This first part offers a comprehensive overview of the CRRD dataset, contextualizing the registry's scope, data providers, and reporting period. It sets the stage for the following chapters, which present:

1. **Introduction:** Explanation of the report structure, objectives, scope and structure of Part I of the report.
2. **Dataset Overview:** Summary of participating centres' reporting activity over time.
3. **Global Data Quality Assessment:** Evaluation of completeness, validity and reliability, consistency, and disease-coding quality using aggregated data quality indicators across all participating centres.
4. **Global Validation Workflow Analysis:** Overview of validation process indicators and disease coding-related validation patterns across all participating centres.
5. **Summary of Key Findings:** A high-level interpretation of strengths and areas where further harmonisation efforts may be beneficial.
6. **Conclusion:** Summary of findings and potential next steps.
7. **References:** Supporting documentation and methodological notes.

2. DATASET OVERVIEW

This section presents the overall volume and evolution of data submissions collected in the CRRD between 2015 and 2025, combining all participating genetic centres under Architecture 1 (v1–v3). It first describes the **registry scope and data collection framework**, including the case definition, participating data providers, and data flow. It then summarises the **composition of the dataset** and **trends in data submissions over time**.

2.1. REGISTRY SCOPE AND DATA COLLECTION FRAMEWORK

2.1.1. CASE DEFINITION AND INCLUSION CRITERIA

The target population of the CRRD comprises all individuals affected by a rare disease whose care pathway involves an accredited genetic centre or RDF in Belgium, either through direct consultation or through the analysis of biological samples by the centre. Once a diagnosis is established, the patient becomes eligible for inclusion in the CRRD.

The case definition for the data collected in the CRRD under Architecture 1 (v1–v3) is as follows:

1. The patient has a diagnosis of a **rare disease**, a **rare genetic disorder**, or a **rare cancer**.
2. The patient is **symptomatic**; asymptomatic individuals or carriers are not eligible.
3. **Registration occurs after birth**, allowing the patient to actively participate in the healthcare system. Prenatal diagnoses, prenatal deaths, and stillbirths are not registered.
4. **Each case** meeting the **above criteria** must be registered.

Additional clarification on the case definition:

- The diagnosis may be established based on clinical assessment and/or confirmed by genetic or laboratory tests.
- Data may be collected during diagnostic evaluation, as well as during follow-up or other clinical visits.
- For disease inventory, classification, and reference, the CRRD relies on [Orphanet](#), the international reference portal on rare diseases and orphan drugs for healthcare professionals and the general public. Inclusion in Orphanet requires, among other criteria, that the disease meets the European definition (i.e., affecting fewer than 5 in 10 000 individuals).
- For the purposes of this report, **registrations** refer to individual submissions to the CRRD, which may include multiple entries for the same patient (e.g., repeated reporting over time, updates, or submissions from different data providers). Unique patient **records** refer to deduplicated patient-level entries, where multiple registrations are linked to a single individual through the CRRD pseudonymisation and linkage procedures.

2.1.2. DATA PROVIDERS

The CRRD was implemented progressively following its migration to Healthdata.be in July 2015. Technical onboarding of the eight national genetic centres occurred between August 2015 and April 2016, marking the start of data submissions under Architecture 1 (v1–v3). All eight centres were expected to contribute data to the registry.

However, the Centre de Génétique Humaine de l'ULB did not submit data to the CRRD due to data-privacy concerns raised by its ethical committee and was therefore not included in this reporting cycle. Consequently, the analysis in this report covered seven of the eight initially foreseen data providers.

Table 1 presents the technical onboarding timeline of the genetic centres in chronological order of their integration into the Healthdata.be platform.

Table 1. Technical onboarding of the eight accredited genetic centres.

Data providers	Technical onboarding
UZ Antwerpen – Centrum Medische Genetica	18/08/2015
UZ Gent – Centrum voor Medische Genetica	16/09/2015
ULB – Centre de Génétique Humaine	21/10/2015
Institut de Pathologie et de Génétique	30/10/2015
CHU de Liège – Centre de Génétique Humaine	3/12/2015
Cliniques universitaires Saint-Luc – Centre de Génétique Humaine	17/12/2015
UZ Brussel – Centrum voor Medische Genetica	14/01/2016
UZ Leuven – Centrum Menselijke Erfelijkheid	20/04/2016

2.1.3. DATA FLOW

Data processing within Architecture 1 follows the standardised workflow developed by Healthdata.be, ensuring secure data collection, encryption, validation and analysis across all participating centres.

Data providers collect information directly from their local operational systems (e.g., Electronic Health Records (EHR), Laboratory Information Management System (LIMS) or Health Management Information System (HMIS)). These data are then made available either by automatically retrieving Comma-Separated Values (CSV) files or through manual input, and are transmitted securely to Healthdata.be via the eHealthBox, the national encrypted mailbox system managed by the eHealth platform.

Transmission occurs either through local system integration or via the Healthdata for Data Providers (HD4DP) software component, installed and managed locally by each centre. HD4DP connects to the central Metadata catalogue (Data Collection Definition, DCD), which contains the project-specific variable lists, value sets, and validation rules published by Healthdata.be³.

Before delivery to Healthdata.be, the eHealth platform pseudonymises and encrypts personal identifiers, such as the Belgian National Registration Number (NISS). A subsequent pseudonymisation step is applied within Healthdata.be to ensure alignment between legacy and newly collected data and to support longitudinal analyses under a uniform pseudonymisation scheme⁴. The encrypted datasets are then forwarded to Healthdata.be, where automatic receipt confirmation is sent back to the data provider.

Sciensano's researchers and data managers access the submitted datasets through the Healthdata for Researchers (HD4RES) platform. Within HD4RES, they monitor submission status, perform initial data validation checks, and issue feedback requests to the data providers when corrections are required. After validation and prior to any analytical processing, a Small Cell Risk Assessment (SCRA) is performed to ensure compliance with legal, ethical and data protection requirements. Only after these steps are completed are the validated datasets integrated into the CRRD central database for analysis and reporting⁵.

³ For further details, please consult: <https://docs.healthdata.be/HDBP0008>.

⁴ For further details, please consult: <https://www.ehealth.fgov.be/ehealthplatform/file/cc73d96153bbd5448a56f19d925d05b1379c7f21/8b1b2e5884f4b3bae0045e3f81cfeb5a030faca2/15-009-f102-hd4patient-modifiee-le-3-mars-2020.pdf>.

⁵ For further details, please consult: <https://healthdata.sciensano.be/fr/l-architecture-technique-de-healthdata-be>.

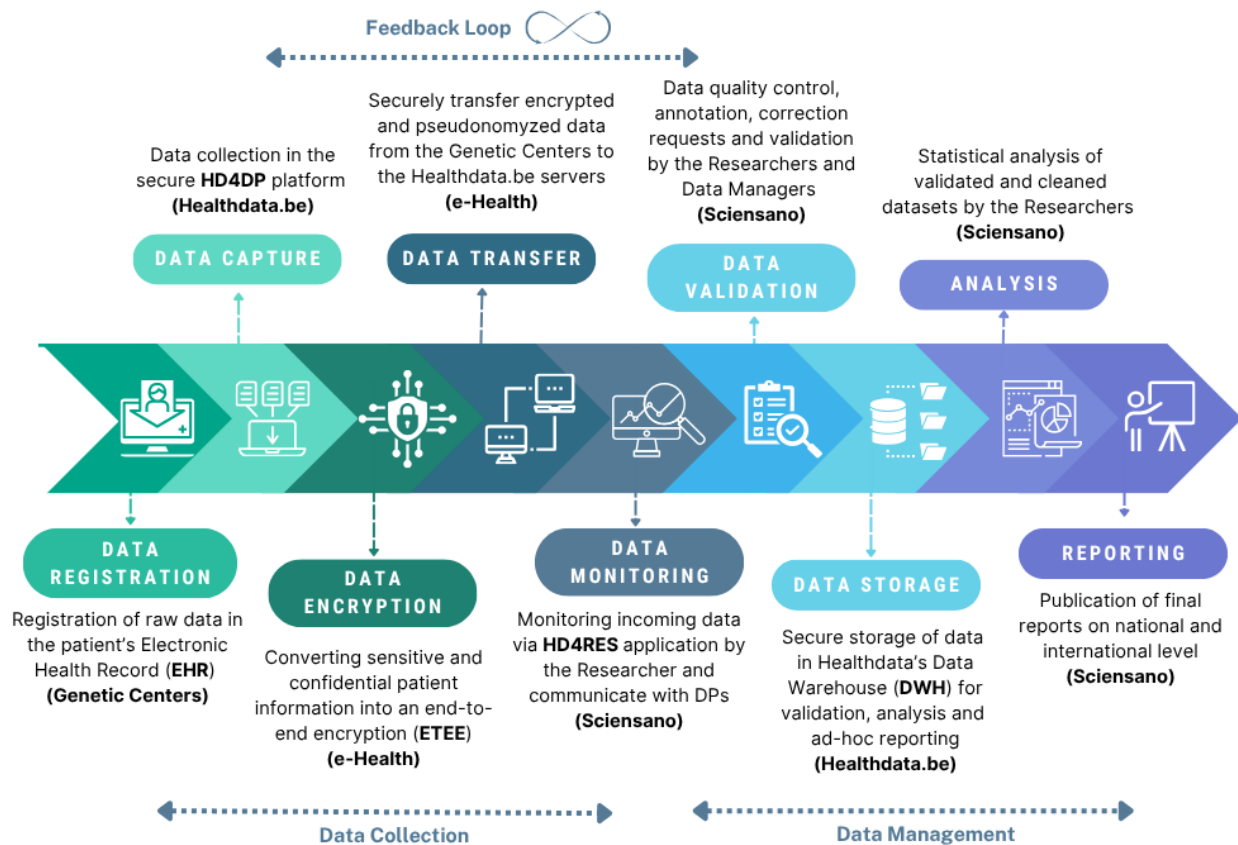


Figure 1. Data flow in the CRRD under Healthdata.be Architecture 1.

Figure 1 illustrates the end-to-end data collection and management workflow for the CRRD, from data capture in the genetic centres to national reporting by Sciensano. It highlights key stages; registration, encryption, monitoring, validation, storage and analysis, supported by a continuous feedback loop between the CRRD management team and the data providers.

Data validation within the CRRD follows a two-step process. First, **front-end validation** is performed at the level of the data provider through the HD4DP component. At this stage, mandatory fields, format rules and predefined value sets are automatically checked, preventing structurally invalid records from being submitted to the central platform. Second, **back-end validation** is conducted by the CRRD management team within the HD4RES environment. These checks focus on the internal consistency and plausibility of submitted data. When inconsistencies or errors are detected, feedback requests are sent to the data providers, who may correct and resubmit the records. This iterative validation process can result in multiple submissions for the same patient record.

A detailed analysis of the validation workflow and its associated indicators is presented in Chapter 4: Global Validation Workflow.

2.2. DATASET COMPOSITION AND METADATA EVOLUTION

This section summarises the **variables collected in the CRRD under Architecture 1 (v1–v3) and the evolution of the registry's metadata**, as defined in successive versions of the DCD.

The evolution of the CRRD's metadata is the result of an iterative development process that began with a Proof of Concept (POC) phase conducted between December 2013 and June 2015 with two pilot centres (i.e., genetic centres affiliated with CHU de Liège and UZ Gent). This phase allowed the first technical tests of the Healthdata.be infrastructure. Early feedback and analysis of the POC data supported the optimisation of both the dataset and submission procedures, which were later integrated into updates of the DCD.

Following this pilot, the technical onboarding of additional centres began in 2015, and most started regular data collection in 2016. Feedback from the initial implementation of **CRRD v1.2** led to several improvements in data structure and feasibility. These revisions were validated by the Steering Committee (2016) and the Belgian College of Human Genetics (2017), resulting in the release of **CRRD v2.0**. A subsequent update, **CRRD v3.0**, was launched late 2018 to address remaining technical issues and further optimise the data collection process. Table 2 summarises the key variable changes introduced across the three versions.

Table 2. Overview of variables collected in the CRRD (v1.2 – v3.0).

Variable	Mandatory variable in DCD v1.2	Mandatory variable in DCD v2.0	Mandatory variable in DCD v3.0
Identification			
Unique NIHDI-code of the registering centre	yes	yes	yes
NIHDI-code of the specialist (pseudonymised)	yes	yes	yes
Patient ID (pseudonymised)	yes	yes	yes
Date of birth (SCRA)	yes	yes	yes
Sex	yes	yes	yes
Place of residence	yes	yes	yes
Deceased	not present	no	no
Date of death*	no	no	no
Country of residence	yes	yes	yes
Consultation			
Referred to registering centre by	not present	yes	yes
Country of referral	not present	yes	yes
Type of service requested	not present	no	yes
Date of first consultation in the registering centre	yes	yes	yes
Time of current diagnosis	no	no	yes
Date of current consultation	not present	yes	yes
Country of current consultation	not present	yes	yes
City of current consultation	not present	yes	yes
Diagnosis			
Time of first symptoms (first consultation)	no	no	yes
Status diagnosis	yes	yes	yes
Base of diagnosis	yes	yes	yes
Diagnosis*: ORPHAcode or other code (i.e. ICD-10, SNOMED CT, ICD-O, HPO, OMIM, ISCN, HGVS)	min. 1 coding system required	min. 1 coding system required	min. 1 coding system required
Diagnosis: Disease name	not present	yes	yes
Referred from registering centre to	not present	no	yes
Other			
Consent of patient for facilitation of recruitment	yes	no	no

Note: For further details on the variables, including their corresponding answer options, see <https://dcd.healthdata.be/en/collection/CRRD/>

Additional clarification on the variables (*):

- “Date of death” is a conditional variable, displayed only when the preceding variable “Deceased” is completed as “Yes”.

- “*Diagnosis*” uses multiple disease classification systems; although no specific system was mandated, preference was given to the Orphanet nomenclature, as it is considered the most comprehensive and accurate system for coding rare diseases⁶.

2.3. REPORTING TRENDS

The reporting period spans July 2015–June 2025, corresponding to all registrations submitted and validated under Architecture 1. Over this period, a total of **76 894 registrations** were collected in the CRRD, corresponding to **21 963 unique patient records**.

The difference between total registrations and unique patient records mainly reflects the validation loop built into the CRRD workflow. Centres that received a higher number of correction requests often generate multiple resubmissions for the same patient record, resulting in a higher total registration count. In addition, some centres submitted all patient consultations rather than a single record per patient, which also contributed to inflated total counts in certain years.

Table 3 provides a year-by-year summary of the total number of registrations, along with the unique and validated patient records submitted to the CRRD during this period.

Table 3. Yearly total registrations and unique patient records submitted to the CRRD (2015–2025).

Year	Total registrations (N)	Unique patients (N)	Unique patients (%)
2015	1	1	100,0
2016	10 802	374	3,5
2017	2 666	1 307	49,0
2018	3 792	706	18,6
2019	26 864	3 579	13,3
2020	9 521	4 737	49,8
2021	2 320	949	40,9
2022	10 415	5 272	50,6
2023	4 070	2 793	68,6
2024	1 496	1 304	87,2
2025	4 947	941	19,0
Total	76 894	21 963	28,6

⁶ DCD v1–v3 help text extract: "A disease can be described by using different coding systems. **Preference is given to the use of the Orphanet classification because it is the most complete and accurate classification system for rare diseases.** If multiple coding systems are used to describe the disease, different codes can be inserted in the respective fields. It is mandatory to fill in at least one of these fields..."

To visualise the evolution of yearly submissions, Figure 2 presents the cumulative number of total registrations and unique patient records per year, illustrating the registry’s overall growth trajectory between 2015 and 2025.

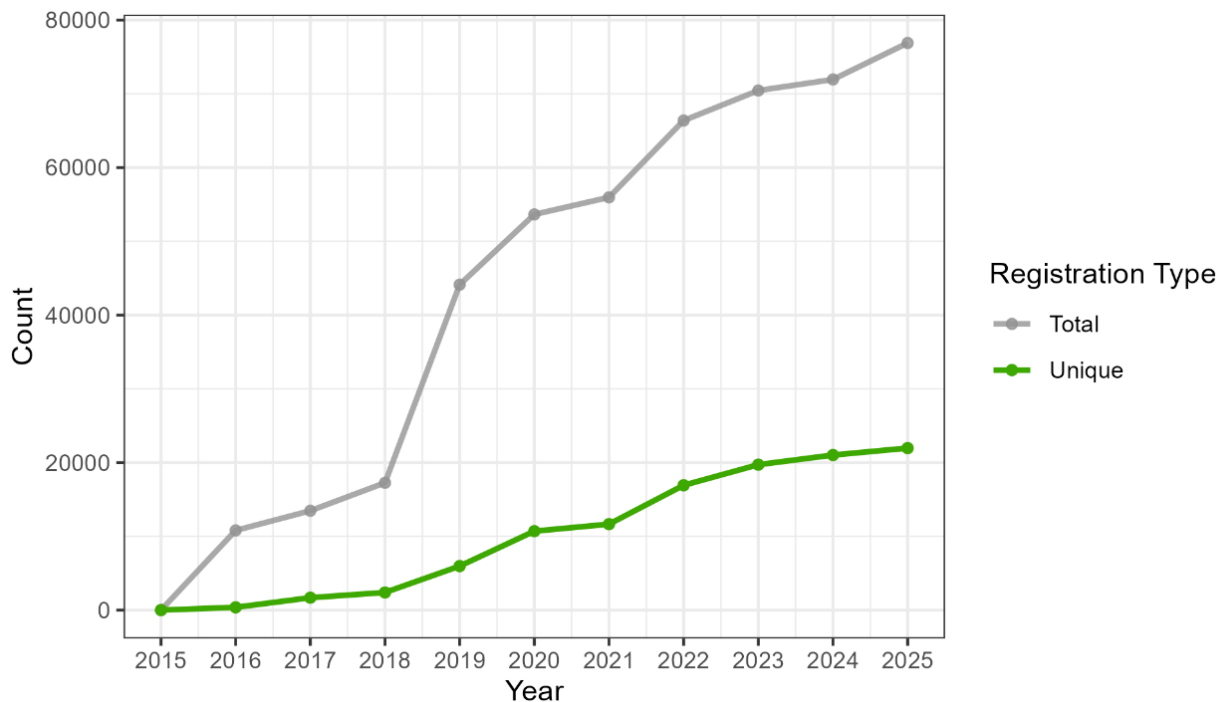


Figure 2. Evolution of yearly unique patient records submitted to the CRRD (2015–2025).

Figure 2 illustrates the **progressive increase** in yearly cumulative total registrations and unique patient records under Architecture 1 (2015–2025), reflecting both the gradual onboarding of genetic centres to the registry (see Table 1) and continued registration efforts over time.

A first substantial increase in the cumulative total number of registrations was observed in **2016**, when all genetic centres had onboarded. A second marked increase occurred in **2019**, following the release of CRRD v3 in 2018, which introduced a more stable data structure and facilitated large-scale uploads from participating centres. The subsequent drop **between 2020 and 2022** may partly reflect the impact of the COVID-19 pandemic on clinical activity, data entry, and validation workflows. Fluctuations between **2023 and 2024** likely reflect system transitions prior to the migration to Architecture 2 in 2024, both at the level of Healthdata.be and among data providers.

Data for **2025** are included up to June 2025 only, corresponding to the official closure of Architecture 1; as such, figures for 2025 represent partial-year data and should be interpreted with caution.

3. GLOBAL DATA QUALITY ASSESSMENT

The following chapter provides an **aggregated assessment of overall data quality within the CRRD**, based on the full set of 76 894 registrations submitted under Architecture 1 (v1–v3). The analysis evaluates key data quality dimensions (i.e., completeness, validity and reliability, consistency and disease-coding quality) across the registry as a whole, independently of individual data providers.

The purpose of this global assessment is to evaluate how reliably and consistently data submitted to the CRRD reflect information captured at source level, to identify recurring or systemic quality issues, and to support continuous improvement of data entry processes, validation rules, and technical specifications.

3.1. METHODOLOGY

The dataset used for this assessment included **all registrations submitted under Architecture 1, irrespective of their validation status**.

All indicators presented in this chapter are derived from standardised data quality programs implemented in SAS Enterprise Guide. These rule-based checks detect missing values, logical inconsistencies, format errors, and incorrect or incomplete use of disease coding systems (e.g., ORPHAcodes, ICD-10, SNOMED CT). These programs evaluate four main data quality dimensions; **completeness, validity and reliability, consistency and disease-coding quality**, based on predefined rules and thresholds. Each dimension is summarised in Table 4.

Table 4. Overview of data quality dimensions, definitions and assessment approaches across the CRRD.

Data quality Dimension	Definition	Assessment approach
Completeness	Extent to which mandatory variables contain non-missing values	Percentage of registrations with non-missing data per variable or domain
Validity and reliability	Degree to which variable values comply with predefined formats, reference lists or value ranges	Automated format and range checks, validation against reference tables (e.g., country codes, postcodes, date logic)
Consistency	Logical coherence between related variables within a record	Cross-field checks to flag contradictions or illogical combinations
Disease-coding quality	Accuracy and completeness of disease coding using standardised disease classification systems	Verification of valid codes and alignment between name-code pairs

All outputs are aggregated across participating centres to provide a **global-level overview**, enabling detection of systemic quality issues rather than centre-specific deviations. Detailed results per centre are presented in the centre-specific benchmark reports (Part II).

3.2. COMPLETENESS

3.2.1. MISSING VALUES

Completeness indicators quantify the proportion of mandatory variables containing valid, non-missing data across the entire registry. It provides an overall view of how thoroughly the information required for each patient record has been captured by the participating genetic centres.

The analysis was performed on all mandatory variables from the Identification, Consultation and Diagnosis domains, with the criterion for inclusion being “mandatory in at least one version” (see Table 5).

Table 5. Mean completeness of mandatory variables per domain across CRRD registrations (%).

Domain	Mean completeness (%)	Most frequent missing variable
Identification	100,0	Place of residence
Consultation	93,2	Time of current diagnosis
Diagnosis	76,7	Referred from the registering centre to

Note: Completeness is calculated as the percentage of registrations with non-missing values for each variable within the domain. Percentages are rounded to one decimal place. The reported 100,0% completeness for the Identification domain corresponds to an underlying value of 99,99867%, with the residual incompleteness attributable to missing postcode values.

Overall, completeness levels were highest for Identification variables, reflecting their mandatory status across all CRRD versions. Lower levels were observed in the Consultation and Diagnosis domains, where certain variables were introduced or reclassified as mandatory only in later DCD versions (v3.0).

To complement the aggregated results by variable domain, Table 6 details the completeness percentage for each individual variable included in the CRRD.

In the **Identification domain**, all mandatory variables achieved 100,0% completeness, suggesting that this information was readily captured from EHRs. This contrasts with other domains, where lower completeness levels likely reflect greater complexity of the variables and challenges in extracting or retrospectively completing the required information.

Within the **Consultation domain**, the variables “*Time of current diagnosis*” and “*Time of current diagnosis: specify*” had lower completeness levels (82,5% and 86,8%, respectively). This pattern is likely linked to difficulties in capturing this information from EHRs, often requiring manual completion or resubmission through the validation feedback loop. The lower completeness for the variable “*Type of service requested*” was expected (84,7%), as it was only mandatory in DCD v3.0. By contrast, the variable “*Referred to registering centre by*” exhibited a similarly reduced completeness level despite being mandatory across all DCD versions. This discrepancy is likely linked to a front-end validation bug present during part of the reporting period, whereby the field was marked as mandatory in HD4DP1 but did not effectively block submission when left empty. As a result, incomplete values could be accepted despite the defined validation rule.

In the **Diagnosis domain**, the variable “*ORPHAcode*” displayed a substantially lower completeness (68,4%). This result can be explained by a methodological consideration: during the reporting period, registrants were required to provide at least one disease code, without specifying the coding system to be used. Completeness is nevertheless reported specifically for ORPHAcodes in this section, given the focus on this classification system in subsequent analyses. The very low completeness observed for “*Referred from registering centre to*” was expected, as this variable was only introduced as mandatory in DCD v3.0. The variable “*Time of first symptoms (first consultation)*” achieved a completeness level of 84,6%, likely reflecting the inherent difficulty of capturing this information clinically and/or extracting it reliably from EHRs.

Table 6. Mean completeness of variables across CRRD registrations (%).

Variable	Mandatory variable in all DCD versions	Mean completeness (%)
Identification		
Country of residence	yes	100,0
Date of birth	yes	100,0
Place of residence	yes	100,0
Sex	yes	100,0
Consultation		
Country of referral	yes	100,0
City of current consultation	yes	100,0
Country of current consultation	yes	100,0
Referred to registering centre by	yes	84,7
Time of current diagnosis	no ^a	82,5
Type of service requested	no ^b	84,7
Date of current consultation	yes	100,0
Date of first consultation in the registering centre	yes	100,0
Time of current diagnosis: specify	no ^{a,c}	86,8
Diagnosis		
Base of diagnosis	yes	98,3
Status diagnosis	yes	100,0
ORPHAcode	min. 1 coding system required	68,4
Time of first symptoms (first consultation)	no ^a	84,6
Diagnosis: Disease name	yes	100,0
Referred from registering centre to	no ^b	9,0

^a Non-mandatory in v1.2 and 2.0; mandatory in v3.0.

^b Not present in v1.2; non-mandatory in v2.0; mandatory in v3.0.

^c Conditional variable, applicable only for specific values of the previous variable "Time of current diagnosis".

Note: Completeness is calculated as the percentage of registrations with non-missing values for each variable.

Variables with completeness levels below aggregated mean were further examined in the individual benchmark reports prepared for each participating centre. These reports provide targeted feedback and recommendations to support the harmonization and continuous improvements of data registration practices across the CRRD.

3.2.2. USE OF “UNKNOWN” VALUES

In addition to missing data, the completeness assessment also considered the use of the “unknown” value in variables where this option is explicitly provided in the DCD. While “unknown” entries are not technically missing, their frequency offers valuable insight into the availability within the EHR and quality of clinical information at the time of registration.

Table 7 summarises the proportion of “unknown” values recorded for key variables across the Identification, Consultation and Diagnosis domains.

Table 7. Total percentage of “unknown” values per selected variable across CRRD registrations (%).

Domain	Variable	Total percentage across CRRD registrations	Interpretation
Identification	Sex	0,03	The very low proportion of ‘unknown’ values indicates that this variable is documented in a near-systematic manner.
Consultation	Referred to registering centre by	0,00	In combination with lower completeness observed for this variable (84,7%, see Table 6), this pattern suggests that, when the information was not available, the field was more often left blank rather than completed using the ‘unknown’ option.
Consultation	Type of service requested	0,00	In combination with lower completeness observed for this variable (84,7%, see Table 6), this pattern suggests that, when the information was not available, the field was more often left blank rather than completed using the ‘unknown’ option.
Consultation	Time of current diagnosis	36,52	In combination with lower completeness observed for this variable (82,5%, see Table 6), the frequent use of the ‘unknown’ option suggests difficulties in capturing this clinical information, likely due to limited availability or retrospective uncertainty in the EHR.
Diagnosis	Time of first symptoms (first consultation)	19,21	In combination with lower completeness observed for this variable (84,6%, see Table 6), the frequent use of the ‘unknown’ option suggests difficulties in capturing this clinical information, likely due to limited availability or retrospective uncertainty in the EHR.
Diagnosis	Base of diagnosis	0,00	In combination with higher completeness observed for this variable (98,3%, see Table 6), this pattern suggests that the information is generally readily available and systematically captured in the EHR.
Diagnosis	Referred from the registering centre to	0,00	In combination with low completeness observed for this variable (9,0%, see Table 6), this pattern suggests that, when the information was not available, the field was more often left blank rather than completed using the ‘unknown’ option.

Although “unknown” is a valid and permitted value, a high proportion may indicate incomplete documentation or variability in data entry practices between centres. In addition, differences in data availability and structuring across EHR software systems may influence the frequency with which “unknown” is selected for certain variables. Therefore, while these values were treated as valid for completeness calculations, they were interpreted separately from fully documented entries.

To improve data interpretability, centres are encouraged to use the “unknown” option only when information is genuinely unavailable, rather than as a default or placeholder. Monitoring the frequency of “unknown” values over time can help identify areas where clinical documentation or data entry guidance may need reinforcement. In parallel, periodic revisions of the dataset, in alignment with information routinely available in EHR systems, can further support consistent and meaningful data capture across centres.

3.3. VALIDITY AND RELIABILITY

Following the assessment of completeness, this section examines the **validity** and **reliability** of data recorded in the CRRD using data quality indicators.

Validity checks assess whether recorded values **comply with predefined technical and semantic constraints**. They address the question “Is the value allowed?” and verify adherence to expected formats, reference lists, and logical value ranges. Examples include dates recorded in valid formats, age values within plausible bounds, or disease codes belonging to standardised coding systems.

Reliability checks assess whether the recorded information is **meaningful and plausible** from a clinical and epidemiological perspective. They address the question “Does the information make sense?” and focus on the overall coherence of values, including the presence of essential diagnostic information and the alignment between descriptive and coded data elements. Examples include the presence of a disease name and/or code for a diagnosis, or the plausibility of reported age at diagnosis in relation to the clinical context.

Together, these quality controls during front-end and back-end validation help identify structural errors (e.g., invalid dates or codes), logical inconsistencies between related variables (e.g., date of birth occurring after date of death), and implausible values or value combinations. Such checks are essential to ensure that CRRD data are not only complete but also accurate and suitable for aggregated analyses and benchmarking.

A summary of the validity and reliability checks applied to the CRRD dataset is presented in Table 8. It provides an overview of the main variable domains assessed, the types of errors detected and the proportion of affected registrations across all submissions under Architecture 1 (v1–v3).

Table 8. Summary of validity and reliability indicators across key variable domains (%).

Domain	Type of check	Data quality indicator	Total percentage of invalid or unreliable registrations across the CRRD (%)
Identification	Reliability	Birth date after death date	0,00
Identification	Validity	Invalid postcode (not in reference list)	1,50
Identification	Validity	Invalid country code (not in reference list)	0,00
Consultation	Reliability	First consultation before date of birth	0,21
Consultation	Reliability	First consultation after date of death	0,11
Consultation	Reliability	First consultation after current consultation	0,34

Consultation	Reliability	First consultation in the future	0,05
Consultation	Reliability	Current consultation in the future	0,03
Consultation	Validity	Invalid full date format	0,68
Consultation	Validity	Invalid year format	0,02
Consultation	Validity	Invalid month-year format	0,02
Consultation	Validity	Invalid age-range format	0,26
Consultation	Validity	Invalid gestational-age format	0,08
Consultation	Validity	Invalid country code (current consultation)	0,00
Consultation	Validity	Invalid city/postcode (current consultation)	0,70
Diagnosis	Validity	Invalid full date format	0,98
Diagnosis	Validity	Invalid year format	0,00
Diagnosis	Validity	Invalid month-year format	0,01
Diagnosis	Validity	Invalid age-range format	1,24
Diagnosis	Validity	Invalid gestational-age format	0,42
Diagnosis	Reliability	No disease name nor disease code available	0,00
Diagnosis	Reliability	Disease code available but no disease name	0,00
Diagnosis	Reliability	Disease name available but no disease code	2,45
Diagnosis	Validity	Invalid ORPHAcode	0,00
Diagnosis	Validity	Invalid HPO code	0,00
Diagnosis	Validity	Invalid ICD-10 code	93,92
Diagnosis	Validity	Invalid ISCN code	100,00
Diagnosis	Validity	Invalid OMIM code	0,00
Diagnosis	Validity	Invalid SNOMED CT code	0,00

Note: Percentages are calculated using the number of registrations where the relevant variable was recorded as the denominator. For coding system validity checks (e.g., ORPHAcodes, SNOMED CT), the denominator corresponds to the total number of CRRD registrations in which the respective code was recorded.

Table 8 indicates that, within the Diagnosis domain, most registrations **contained both a disease name and at least one corresponding disease code**. Overall, 97,55% of registrations included a disease name together with at least one disease code.

However, 2,45% of registrations contained a **disease name but no corresponding disease code**. This pattern reflects cases where descriptive disease information was provided without the associated coded disease identifier.

Given that the presence of at least one disease code is defined as mandatory, these patterns suggest a **front-end validation limitation** during part of the reporting period, whereby submissions could be sent despite incomplete disease coding. This behaviour points to a system-level validation issue rather than systematic data entry omissions by reporting centres.

Notably, **no invalid ORPHA codes were identified**, indicating that when ORPHA codes were provided, they generally complied with the expected format and reference standards.

Additionally, a high proportion of **ICD-10 codes** (93,92%) and **all ISCN codes** (100,00%) were affected by validity errors. These elevated error rates likely reflect structural aspects of data capture rather than systematic data entry issues.

For **ICD-10**, although codes are expected to conform to a predefined reference list, instances were observed in which free-text disease descriptions were submitted without a corresponding ICD-10 code. This pattern may partly reflect a lack of front-end validation configurations of the HD4DP1 web-based form during the reporting period, which did not consistently enforce format or reference-list validation at submission.

In the case of **ISCN**, codes were recorded exclusively as free-text entries under Architecture 1 and were therefore inherently more susceptible to format-related validity errors. Consistently, free-text disease descriptions without a corresponding ISCN code were also observed, further contributing to the high proportion of detected validity issues for this variable.

Postcodes did not conform to the reference list in 1,50% of all registrations. This pattern likely reflects a combination of factors, including incomplete registrations, front-end validation limitations in the HD4DP1 form, and the use of legacy, temporary, or non-residential postcodes, rather than the systematic use of invalid postcodes.

Invalid **age-range format** was observed in 1,24% of all registrations. This may reflect variability in the interpretation of the form's help text, together with the absence of enforced format front-end validation rules in the HD4DP1 form during the reporting period.

For **all other assessed variables**, validity or reliability errors were observed in **fewer than 1%** of registrations.

3.4. CONSISTENCY

Consistency checks evaluate the logical coherence of data recorded within and across related fields in the CRRD. While validity checks confirm that individual variables meet expected formats and value ranges, consistency checks focus on whether combinations of variables make sense together. This dimension of data quality ensures that the recorded data reflect realistic and non-contradictory clinical situations. Typical consistency rules include, for example, verifying that the referral country corresponds to the country of the reporting centre, or that the basis of diagnosis aligns with the presence or absence of reported symptoms.

An overview of the consistency rules applied across the registry is presented in Table 9.

Table 9. Summary of cross-field consistency indicators across the CRRD (%).

Domain	Data quality indicator	Total percentage of inconsistent registrations across the CRRD (%)
Identification	Missing date of death (among deceased)	0,06
Identification	Postcode = 9999 but country missing	0,00
Identification	Postcode = 9999 but country = BE	0,01

Identification	Postcode = 999 but country filled	0,12
Consultation	Country = BE but city = “not in Belgium”	0,00
Consultation	Country = BE but city missing	0,00
Consultation	Missing country but city filled unexpectedly	0,00
Diagnosis	No-symptoms but time of first symptoms filled	0,00
Diagnosis	No-symptoms but clinical signs coded	0,00
Diagnosis	Basis of diagnosis is “Other” but content is missing	0,27

Note: Placeholder code ‘9999’ should be entered if the patient does not live in Belgium, together with the country of residence. Placeholder code ‘999’ refers to missing data.

Table 9 shows that cross-field consistency errors across all assessed variables were either absent or affected **fewer than 1%** of registrations, indicating a high overall level of internal coherence within the CRRD database.

Among deceased patients, the **date of death** was missing in 0,06% registrations. This rare inconsistency may reflect a combination of factors, including absent or delayed updates following initial registration and limitations in front-end validation rules that did not consistently enforce completion of this field at submission.

It should be noted that vital status information is not yet systematically collected within the current registry framework. The CRRD records each patient as a single entry and does not include routine follow-up of vital status over time. As a result, registering centres typically update this information only when correcting or updating other variables, as recording vital status is not mandatory. Consequently, both the completeness and timeliness of vital status data are limited, and the reported figures are likely to underestimate the true number of deceased patients. To address this limitation, a linkage procedure with the National Registry is currently under development, which will enable systematic capture of vital status and date of death, where applicable.

Furthermore, a small number of consistency issues were also observed for **postcodes** used to indicate place of residence, particularly the inappropriate use of “999” and “9999” in combination with completed country or city fields. These patterns reflect semantic inconsistencies in the use of placeholder codes, rather than incorrect postcodes, and remained limited in frequency.

Finally, 0,27% of registrations recorded the **diagnostic basis as “other”** without further specification. This observation is consistent with the fact that specification of the “other” category is not defined as mandatory.

3.5. DISEASE CODING QUALITY

3.5.1. GENERAL OVERVIEW

The disease-coding quality assessment examines the use and validity of standardised disease classification systems in the CRRD. Accurate and harmonised coding is essential to ensure the interoperability of rare disease data both on a national and international level.

Since its implementation, the CRRD has supported several coding systems, with ORPHAcodes serving as the primary classification standard and international reference for rare diseases, as recommended by the European Commission.

Other coding systems and ontologies (i.e., ICD-10, SNOMED CT, ICD-O, HPO, OMIM, ISCN, LOINC) are used as complementary references to enhance interoperability and facilitate mapping with clinical, epidemiological and genetic datasets, thereby capturing additional phenotypic and genetic information.

Table 10 summarises the distribution of disease coding systems and ontologies used across the registry, illustrating their evolution across the 2015–2025 reporting period. Coverage percentages are calculated as **weighted averages across centres**, using the number of registrations across centres and year as weights. As a result, centres contributing larger volumes have a proportionally greater influence on the aggregated global estimates. Percentages are calculated independently for each coding system and are therefore **not mutually exclusive**, as multiple coding systems may be recorded within a single registration.

Table 10. Distribution of disease coding systems and ontologies used in the CRRD.

Year	ORPHA code coverage (%)	SNOMED CT coverage (%)	ICD-10 coverage (%)	HPO coverage (%)	OMIM coverage (%)	ISCN coverage (%)	No code (%)
2015	100,00	0,00	0,00	0,00	0,00	0,00	0,00
2016	97,29	0,00	5,34	2,22	1,18	0,00	0,00
2017	98,53	0,00	5,88	0,00	0,48	0,00	0,00
2018	78,44	20,83	0,50	0,00	0,76	0,00	0,00
2019	83,14	14,19	0,00	7,78	1,83	0,03	0,14
2020	76,44	22,11	0,00	0,23	1,34	0,00	0,52
2021	72,60	25,82	0,00	0,00	0,09	0,00	1,50
2022	81,48	5,70	0,00	6,86	10,92	0,00	4,42
2023	58,26	9,75	0,05	0,00	1,41	0,00	31,02
2024	99,60	0,00	0,00	0,00	0,40	0,00	0,40
2025	80,61	2,99	0,00	62,71	19,28	0,07	0,70

For all registrations submitted under Architecture 1 (2015–2025), **ORPHA**codes remained the dominant disease classification system throughout the reporting period, reflecting their role as the international reference standard for rare diseases. It should be noted, however, that this reflects usage within the **registry context only**. While some centres use ORPHAcodes routinely in their EHRs, others apply them exclusively for CRRD reporting purposes or rely on manual transcoding to comply with registry recommendations. In addition, registrations submitted without an ORPHAcode were returned to centres through the validation feedback loop whenever completion is feasible, which may further contribute to the high observed coverage of ORPHAcodes in the CRRD dataset.

Other disease coding systems and ontologies were also used to submit registrations to the CRRD, either alone or alongside ORPHAcodes. **SNOMED CT** coverage increased progressively and peaked at 25,8% in 2021, reflecting its growing role as a core clinical terminology.

OMIM coverage peaked at 10,9% in 2022, while **HPO** coverage reached a maximum at 62,7% in 2025. This high aggregated value reflects concentrated HPO usage within at least one high-volume centre, rather than uniform adoption across all centres. Unlike disease classification systems, ontologies such as **OMIM** and **HPO** are designed to describe genetic and phenotypic characteristics and are therefore **commonly applied multiple times within a single registration**. For example, an OMIM entry for a clinical diagnosis can reference several associated genes and phenotypes, while HPO enables a patient’s phenotypic profile to be captured using multiple terms describing clinical features, severity, and onset. As such, the presence of multiple OMIM or HPO terms within a single patient record is expected and reflects the appropriate and intended use of these ontologies.

ICD-10 coverage remained limited throughout the reporting period and peaked at 5,9% in 2017, after which no further ICD-10-coded registrations were submitted. The disappearance of ICD-10 coding in CRRD submissions from 2019 onwards is consistent with the ongoing national transition towards ICD-11, supported by the increased use of SNOMED CT as the core clinical terminology. This transition is expected to be completed by 2027. **ISCN**, the standard for reporting human chromosomal abnormalities, was used only in 2019 and 2025, indicating limited coverage throughout the reporting period.

Overall, these trends illustrate a gradual convergence towards ORPHAcodes as the primary rare disease classification system within the CRRD, alongside evolving use of complementary coding systems reflecting broader national and international developments in data interoperability.

4. GLOBAL VALIDATION WORKFLOW ANALYSIS

This chapter examines the performance of the validation workflow applied to the CRRD submissions under Architecture 1 (v1–v3). While the previous chapter focuses on the quality of all the submitted data, this chapter analyses key indicators related to the CRRD data validation workflow.

All registrations pass through a two-tier workflow within the Healthdata.be infrastructure:

- **Front-end validation** (HD4DP), where mandatory fields, format rules and value sets prevent structurally invalid registrations from being submitted, and;
- **Back-end validation** (HD4RES/SAS), where additional logical, cross-field and disease coding checks are applied after data reception.

This chapter summarises the results of these validation steps, including the distribution of validation process indicators, the most common error types and the resubmissions triggered through the feedback loop. These insights help assess how effectively the validation workflow supports data quality improvement across the registry.

4.1. METHODOLOGY

At the data provider level, the HD4DP application applies front-end validation rules embedded in the electronic data collection form. These rules include mandatory fields, format restrictions and dropdown lists designed to prevent submission of incomplete or structurally invalid registrations. Those that do not meet these criteria cannot be transmitted to the registry.

Once submitted, registrations are processed through a series of back-end validation rules performed by Sciensano's research team via SAS Enterprise Guide. These checks include additional logical, disease coding and cross-field consistency rules that go beyond the front-end validations and are applied to the consolidated dataset after data reception.

This second layer of validation ensures that submitted data comply with CRRD data definitions, align with reference classifications and maintain internal coherence.

Each record is then categorised according to its validation outcome:

- **Validated:** registrations meeting all required format, completeness and consistency criteria, and are therefore stored in the registry's DWH;
- **Correction Needed:** registrations containing incomplete or inconsistent information requiring clarification by the data provider. These registrations are **temporarily rejected** and returned to the data provider for correction via the feedback loop to the HD4DP1 platform. Once corrected and resubmitted, they may subsequently be validated and accepted into the registry's DWH;
- **Rejected:** registrations that are **definitively rejected** due to critical structural or content-related errors, including duplicate entries or registrations not meeting the CRRD inclusion criteria. These registrations are not eligible for analysis and are not incorporated into the registry's DWH.

Registrations flagged as "Correction Needed" are returned to data providers via Architecture 1's feedback loop, enabling iterative review and resubmission. Each resubmission is treated as a new incoming registration and is counted separately in the total number of registrations. As a result, the number of submitted registrations exceeds the number of unique patients included in the CRRD.

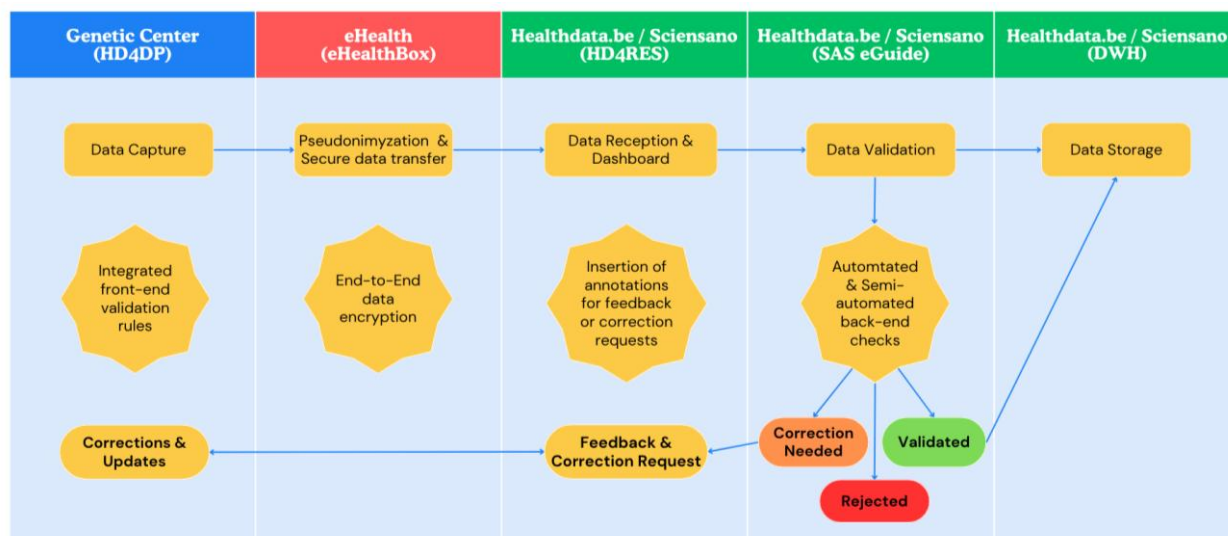


Figure 3. Overview of the two-tier validation workflow applied in the CRRD (Architecture 1).

Figure 3 illustrates the sequential and iterative validation workflow applied through the Healthdata.be infrastructure for CRRD data submissions under Architecture 1. The process combines front-end validation rules at the data provider level (HD4DP) with back-end validation by the CRRD research team (HD4RES). This two-level validation system, combining preventive controls at submission and analytical checks post-submission, aims to ensure the progressive improvement and harmonization of data quality across all participating centres.

4.2. VALIDATION PROCESS INDICATORS

This section presents the distribution and evolution of validation process indicators across all data submissions from 2015 to 2025, providing an overview of how registry data quality has evolved throughout the reporting period. It summarises both the global validation trends (reflecting all submitted registrations) and the disease coding related validation patterns, which represent a major source of feedback and correction activity within the CRRD.

4.2.1. OVERALL VALIDATION TRENDS

Table 11 summarises the annual evolution of validation process indicators, including the total number of registrations, the number of unique patients represented, the average number of registrations per patient and the proportion of first-attempt and multi-attempt validations.

These indicators reflect the iterative nature of the validation workflow, whereby data providers progressively improve their data quality through feedback and resubmissions. They also provide insight into the operational workload for the data providers, as each validation error generates additional steps for review, correction and resubmission.

Acknowledging this workload is essential when interpreting validation metrics, as higher number of resubmissions may indicate both initial data quality issues and the sustained effort required by the data providers to resolve them within Architecture 1's feedback loop framework.

Table 11. Annual validation process indicators and overall registration status distribution (2015–2025).

Year	Total registrations (N)	Unique patients (N)	Mean number of registrations per patient N (SD)	First-attempt validations (%)	Multi-attempt validations (%)
2015	1	1	1,0 (NA)	100,0	0,0
2016	10 802	374	28,9 (55,1)	44,7	55,3
2017	2 666	1 307	2,0 (1,9)	95,2	4,8
2018	3 792	707	5,4 (3,5)	92,8	7,2
2019	26 864	3 584	7,5 (3,4)	51,3	48,7
2020	9 521	4 741	2,0 (3,0)	96,9	3,1
2021	2 320	949	2,4 (0,4)	91,7	8,3
2022	10 415	5 273	2,0 (2,3)	95,3	4,7
2023	4 070	2 796	1,5 (0,6)	95,1	4,9
2024	1 496	1 304	1,1 (0,4)	95,3	4,7
2025	4 947	942	5,3 (3,7)	95,4	4,6

Note: The mean number of registrations per patient is derived by averaging centre-level means per year. The reported standard deviation represents the dispersion of these centre-level values across data providers.

The single registration in **2015** reflects activity during the POC phase conducted between December 2013 and June 2015 with two pilot centres (i.e., genetic centres affiliated with CHU de Liège and UZ Gent).

In **2016**, a high average number of registrations per patient (28,9) was observed, along with a relatively high proportion of multi-attempt validations (55,3%). This pattern likely reflects the initial large-scale onboarding phase, during which data providers faced a higher registration workload while becoming familiar with the DCD and the HD4DP1 submission interface.

From **2017 onwards**, the average number of registrations per patient decreased substantially and fluctuated slightly, while the proportion of first-attempt validations increased markedly. This trend suggests progressive improvement in registration efficiency and data quality, likely driven by increased familiarity with registry requirements, iterative feedback through the validation loop, and gradual harmonisation of data entry practices.

An exception to this general trend was observed in **2019**, when 48,7% of registrations required multi-attempt validation. This increase coincided with the implementation of **DCD v3.0**, which introduced structural and semantic changes to the dataset and required an adaptation period for data providers. The temporary increase in resubmissions therefore reflects a transition phase rather than a deterioration in data quality.

Figure 4 complements Table 11 by illustrating the distribution of first-attempt and multi-attempt validation process indicators across the reporting period, highlighting the overall shift towards higher first-attempt validation rates over time.

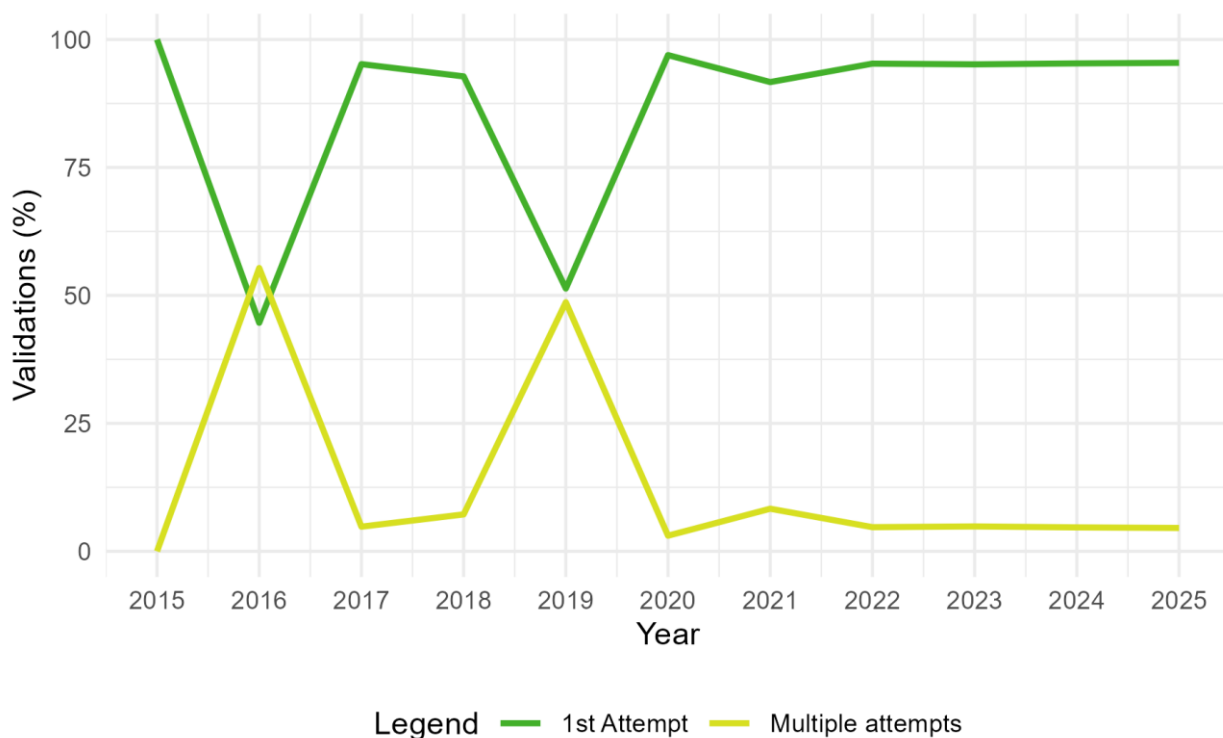


Figure 4. Distribution of first-attempt and multi-attempt validations (2015–2025).

4.2.2. DISEASE CODING RELATED VALIDATION TRENDS

Further investigation of the multi-attempt validations, shows that a substantial proportion of correction requests during the reporting period were linked to disease coding inconsistencies, particularly involving **ORPHAcodes**, which constitute the primary reference system used within the CRRD for rare disease classification and align with international rare disease nomenclature standards.

These corrections most frequently concerned **missing ORPHAcodes, invalid ORPHAcodes, or inconsistencies between the recorded disease name and the associated ORPHAcode.**

Within the validation workflow, registrations were therefore returned to the registering centres whenever the ORPHAcode was missing, invalid, or inconsistent with the recorded disease name, even if another disease code had been provided.

Figure 5 presents the proportion of all validation corrections attributable to ORPHAcode-related issues over time. This allows an assessment of the relative contribution of disease-coding quality specifically related to ORPHAcodes to overall validation corrections (i.e., all corrections, not limited to disease coding) and provides insight into how ORPHAcode-related validation patterns evolved throughout the reporting period.

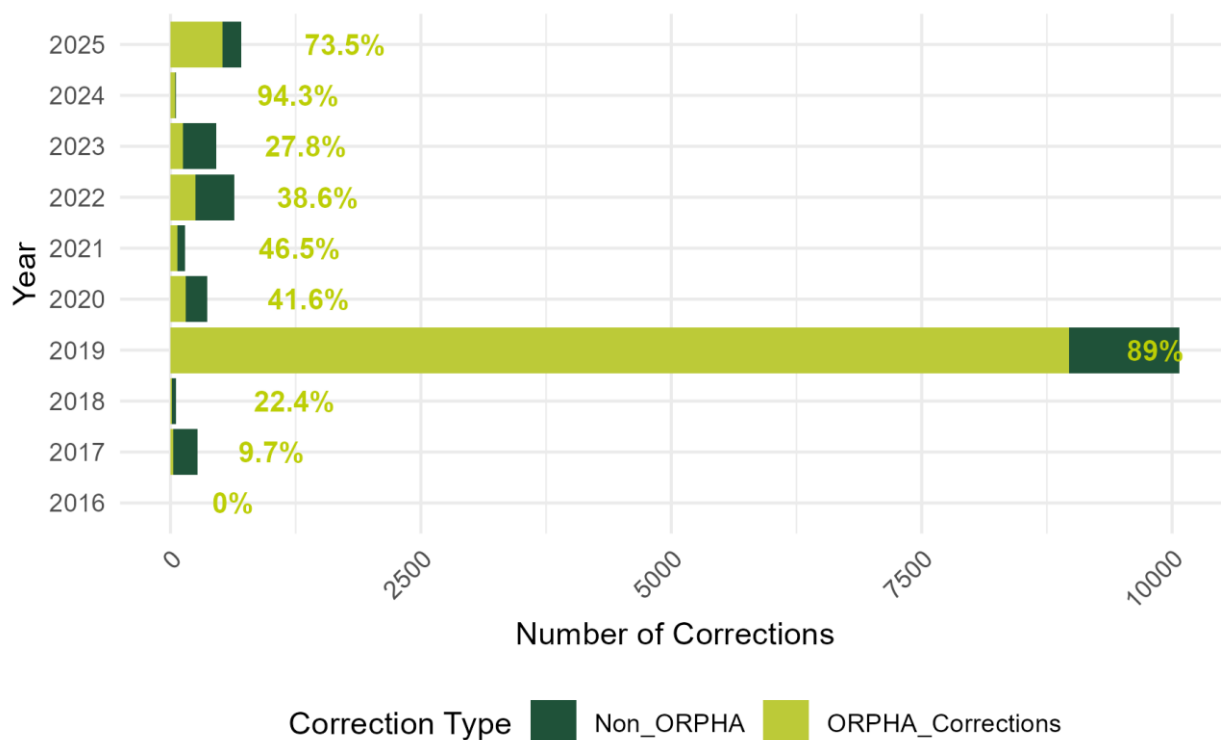


Figure 5. Proportion of validation corrections related to ORPHAcode issues (2015–2025).

Figure 5 highlights a **pronounced increase in the absolute number of validation corrections in 2019**. This peak coincides with a marked rise in total registrations that year (see Table 2), following the release of CRRD v3.0 in 2018, which introduced a more stable data structure and enabled large-scale data submissions from participating centres. The observed surge in corrections therefore reflects both increased registration volume and an expected familiarisation period associated with the introduction of a revised DCD, rather than a deterioration in data quality.

The value of 0% observed for 2016 reflects the very limited number of correction requests recorded during the early implementation phase of the CRRD. Although several multi-attempt validations occurred that year (see Figure 4), only two registrations were ultimately returned to registering centres for correction, and none of these corrections concerned ORPHAcode-related issues. This likely reflects the initial operational phase of the registry, during which validation procedures and feedback loops were still being progressively implemented.

In terms of composition, the **proportion of corrections related to ORPHAcodes was highest in 2019 (89,0%), 2024 (94,3%), and 2025 (73,5%)**. These high proportions reflect the central role of ORPHAcodes in the validation workflow and their systematic verification during the submission process. Importantly, despite the high relative proportion observed in 2024, the absolute number of corrections that year remained very low. Data for 2025 should be interpreted with caution, as they include registrations submitted up to June 2025 only, corresponding to the final months of Architecture 1. The observed patterns for 2025 therefore reflect partial-year activity rather than a full reporting cycle.

Overall, ORPHAcode-related corrections constituted a substantial proportion of validation feedback throughout the reporting period, reflecting both their central role in rare disease classification and the iterative harmonisation process inherent to registry maturation.

5. SUMMARY OF KEY FINDINGS

The analyses presented in Part I of the Data Quality Benchmark Report provide a consolidated global overview of data quality within the CRRD under Architecture 1, drawing on registrations submitted between 2015 and 2025 by the accredited Belgian genetic centres. Taken together, the results demonstrate a maturing registry characterised by generally high data completeness, considerable consistency, and progressively improving validation process indicators over time (e.g., an increasing percentage of first-attempt validations).

At the aggregated-global level, several key strengths emerge. Identification and Consultation-related variables consistently showed high completeness, reflecting the successful capture of core administrative and clinical information across centres. Validity, reliability, and cross-field consistency checks revealed very low error rates for most assessed variables, indicating a high degree of internal coherence and plausibility of the recorded data. In addition, increasing first-attempt validation rates over successive years point to a gradual improvement in registration efficiency and familiarity with registry requirements among data providers.

The analysis also highlights the central role of ORPHAcodes within the CRRD. ORPHAcodes remained the dominant disease classification system throughout the reporting period, and when provided, they generally complied with expected reference standards. Although disease coding accounted for a substantial proportion of validation feedback—particularly during periods of registry transition—these patterns largely reflect the complexity of rare disease classification and the iterative harmonisation process inherent to registry development, rather than persistent data quality deficiencies. In addition, the lack of routine ORPHAcode use in hospital EHRs necessitated retrospective attribution, likely contributing to the high proportion of ORPHAcode-related corrections.

At the same time, the global overview identifies several priority areas for improvement. These include variable completeness in selected diagnostic fields, heterogeneity in disease coding practices across centres, and the operational burden associated with validation and resubmission workflows. Some observed issues were linked to structural or technical characteristics of the data collection framework under Architecture 1, such as limited front-end validation, reliance on free-text fields, and evolving DCDs. These findings underline the importance of aligning variable definitions with data availability in EHRs and strengthening automated validation mechanisms where feasible. Particular attention should be given to the variables “*time of current diagnosis*” and “*time of first symptoms*”, as the findings suggest difficulties in capturing this clinical information, likely due to limited availability or retrospective uncertainty in the EHR.

Importantly, variability observed at the aggregated-global level should be interpreted in context, taking into account differences in local workflows, historical onboarding phases, and transitional periods following updates to registry specifications. Addressing these sources of variability represents an opportunity to further harmonise practices and enhance the comparability of CRRD data.

Against this backdrop, centre-specific benchmark analyses (Part II) provide a complementary perspective by allowing each participating centre to review its own data quality indicators in relation to pseudonymised global trends. Together, the global overview and the centre-level benchmarks are intended to support constructive dialogue, targeted quality-improvement actions, and the continued maturation of the CRRD as a reliable national resource.

6. CONCLUSION

Building on the global synthesis presented in Part I, the findings of this first edition of the CRRD Data Quality Benchmark Report reflect the substantial progress achieved over more than a decade of continuous collaboration between Sciensano, the accredited genetic centres, and the institutional partners involved in the development and operation of the CRRD. Since its inception as a Proof Of Concept initiative, the CRRD is steadily evolving into a national rare disease surveillance system with the capacity to support epidemiological monitoring, data quality improvement, and policy development in the field of rare diseases, while continuing to mature in terms of coverage, relevance, and robustness.

Indeed, the analyses underscore that further improvements remain necessary, highlighting the ongoing work required to fully realise the CRRD's potential and support its long-term sustainability. Enhancing data quality requires coordinated efforts across all components of the data flow, including accurate and complete data entry at centre level; continued refinement of the Data Collection Definition to promote consistent interpretation of variable definitions and alignment with data available in EHRs; refinement of front-end validation rules; sustained investment in technical infrastructure; and strengthened support for data providers. Addressing observed variability in disease coding practices, data completeness, reliability and validity, consistency and validation process indicators will remain essential to further harmonise reporting practices and improve the reliability of national CRRD indicators.

It is important to note that the data quality observed under Architecture 1 benefited from an iterative validation and feedback loop that supported progressive harmonisation of data submission practices within and across centres. Under Architecture 2 (from September 2024 onwards), this feedback mechanism is no longer part of the standard dataflow, placing greater emphasis on data quality at source and upstream practices, rather than on post-submission feedback loops. In line with the evolving national vision for healthcare registries, this shift places greater emphasis on *quality at source* and the proactive availability of accurate, complete, and interpretable data at first submission. Maintaining high data quality therefore increasingly relies on clearly defined data elements, shared standards (e.g., disease coding), interoperable systems, and continued collaboration between all involved parties.

Looking ahead, several ongoing and planned developments offer important opportunities for further improvement. The progressive onboarding of nationally recognised RDFs into the CRRD is expected to broaden the scope of the registry and contribute to more comprehensive capture of the national rare disease population. In parallel, the implementation of the Belgian Plan for Rare Diseases (2026–2030), with its emphasis on standardisation, digital integration, and data interoperability, is anticipated to further strengthen the quality, completeness, and usability of rare disease data.

Together, these efforts are expected to gradually increase national rare disease patient coverage and move the CRRD towards more accurately reflecting the epidemiological landscape in Belgium. Continued collaboration and shared commitment across stakeholders will remain essential to support this progress and to ensure that the CRRD continues to develop as a robust, reliable, and policy-relevant resource for the rare disease community.

7. REFERENCES

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3. For further details, please consult: <https://docs.healthdata.be/HDBP0008>.
4. For further details, please consult: <https://www.ehealth.fgov.be/ehealthplatform/file/cc73d96153bbd5448a56f19d925d05b1379c7f21/8b1b2e5884f4b3bae0045e3f81cfeb5a030faca2/15-009-f102-hd4patient-modifiee-le-3-mars-2020.pdf>.
5. For further details, please consult: <https://healthdata.sciensano.be/fr/l-architecture-technique-de-healthdatabe>.
6. DCD v1– v3 help text extract: "A disease can be described by using different coding systems. **Preference is given to the use of the Orphanet classification because it is the most complete and accurate classification system for rare diseases.** If multiple coding systems are used to describe the disease, different codes can be inserted in the respective fields. It is mandatory to fill in at least one of these fields.

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